Platform: NileConnect Egypt Marketplace Facilitator | **Document Type:** Official Architectural Specification | **Generated:** November 2026

Business Requirements Document (BRD): Egypt Multi-Store Marketplace Facilitator & Outsourced Delivery Platform

1. Executive Summary & Business Model

This Business Requirements Document (BRD) defines the functional, financial, and operational requirements for an **Egyptian Zero-Inventory Multi-Store Marketplace Facilitator** across **Web (Arabic RTL & English)**, **Apple App Store (iOS)**, and **Google Play Store (Android)**.

Core Business Model Pillars

1. **Zero Platform Inventory Risk:** The platform does **not** own warehouses or hold stock. Instead, **Partner Stores (Merchants)** onboard onto the platform, list their products under standardized **Product Categories**, and manage their own live store inventory.

2. **Client-to-Store Connection & Commission Split:** When a Client places an order (even across multiple Partner Stores in a single cart), the platform calculates the **Platform Facilitator Commission (`8%–12%`)** and the **Net Partner Store Payout (`EGP`)**, processing payments via **Paymob Split / Cards / Meeza / Wallets**, **valU BNPL**, **Fawry 48h Reference Codes**, **InstaPay**, or **Cash on Delivery (COD)**.

3. **100% Outsourced Last-Mile Delivery (Store → Client):** Delivery is outsourced to Egyptian 3PL couriers (**Bosta / Mylerz**). Each order automatically splits into **Per-Store Delivery Manifests (`wc_deliveries`)**, instructing the courier to pick up directly from the **Partner Store's branch address** and deliver to the **Client's drop-off address**.

4. **Client Order Monitoring & Historical Closed Archive:** The platform provides clients with full self-service visibility over active in-transit shipments and historical closed orders, 5-stage milestone pipelines, Bosta waybill numbers, printable tax invoices, and 1-click support desk chat.

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2. System Actors & Dedicated Portals

Actor	Role & Responsibilities	Dedicated PoC Screen / Interface
1. Client (Shopper / Consumer)	Filters products by **Product Category** and **Partner Store**, places multi-store orders, pays via Egyptian payment rails, monitors active shipments and historical closed orders, prints official tax invoices, and initiates **Live Customer Support Chats (** `💬 Chat with Support` **)**.	**Tab 1:** `🛒 Client Storefront` **Tab 2:** `📦 My Orders & Tracking (Active & Closed Archive)` **Tab 3:** `📱 iOS & Android Mobile App`
2. Partner Store (Merchant / Vendor)	Manages its own store stock by Category, views incoming store sub-orders, clicks **"Mark Packaged for Bosta Pickup"**, and tracks its **Net Store Payout (EGP)**.	**Tab 4:** `🏪 Partner Store Portal`
3. Outsourced Delivery Coordinator (Bosta / Mylerz 3PL)	Views **Store Pickup Origin → Client Drop-off Destination** manifests, picks up packaged orders from Partner Stores, delivers to Clients, and collects COD cash.	**Tab 5:** `🚚 Outsourced Delivery Hub`
4. Marketplace Facilitator Admin (You)	Has **360° visibility into everything** (All Stores, All Inventory, All Orders, All 3PL Deliveries, All Cash Settlements, All Webhooks), operates the 🎧 **Live Customer Support Desk**, and controls **Commercial Rules (Delivery Fee, Egypt VAT %, Commission %, COD Fee, Cash Settlement Cycle)** across **ALL Partner Stores** or a **Specific Partner Store**.	**Tab 6:** `⚙️ Facilitator Control Tower & Webhooks`

3. Functional Business Requirements (FR)

3.1 Product Categories & Store-Owned Catalog

- **FR-01 (Product Categories Taxonomy):** All products must be organized under bilingual **Product Categories** (`Electronics & Mobile` , `Egyptian Fashion & Cotton` , `Home & Coffee Appliances` , `Beauty & Skincare` , `Sports & Footwear`), with category filter bars on both Web and iOS/Android apps.
- **FR-02 (Partner Store Attribution & Stock Ownership):** Every product card must clearly display its **Category Badge**, **Fulfilling Partner Store Name & Governorate Origin** (e.g., *Cairo Tech Hub — Cairo*), and real-time **Store Available Stock**.
- **FR-03 (Store Inventory Self-Service):** Partner Stores must be able to adjust their stock levels (`+` / `-`) and publish new products under any Category directly from the **Partner Store Portal**.

3.2 Multi-Store Cart, Commission Split & Egyptian Payments

- **FR-04 (Multi-Store Cart & Split Calculation):** When a Client adds products from N different Partner Stores into one cart:
- Each line item calculates `Platform Commission = Line Total × Store Commission %` and `Net Store Payout = Line Total - Platform Commission`.
- Outsourced delivery fee calculates the base Governorate SLA rate (`55–95 EGP`) plus `+25 EGP` per additional store pickup origin.
- **FR-05 (Immediate Partner Store Stock Reservation):** Creating an order immediately reserves the requested units in the respective Partner Store's inventory (`reserved_quantity`).
- **FR-06 (Automatic Stock Return on Fawry Expiry / Decline):** If a 48h **Fawry Reference Code** expires (`FAWRY.EXPIRED`) or a **Paymob** payment is declined, the reserved units are automatically returned to the Partner Store's `available_stock`.

3.3 Outsourced Delivery Hub (Store Pickup → Client Drop-off)

- **FR-07 (Per-Store Delivery Manifest Splitting):** A single Client order containing items from 2 Partner Stores automatically generates **2 separate Outsourced Delivery Manifests (`wc_deliveries`)**, each with its own **Store Pickup Address**, **Client Drop-off Address**, and **Bosta Airway Bill (`BST-xxxxxxx`)**.
- **FR-08 (Store Packaging Handshake):** Partner Stores mark their sub-order `READY_FOR_COURIER_PICKUP` in the **Partner Store Portal**, notifying the **Outsourced Delivery Hub** and triggering a push notification.

3.4 Facilitator 360° Control Tower, Commercial Policy Engine & Customer Support Desk

- **FR-09 (Per-Store & All-Stores Commercial Policy Engine):** The Facilitator Admin must be able to configure **Commission %**, **Egypt VAT % (`14%`)**, **VAT Calculation Mode (`INCLUSIVE` vs. `ADDITIVE`)**, **Store Delivery Fee Override (`EGP` or `0 EGP` Free Shipping)**, **Extra Multi-Store Pickup Fee (`EGP`)**, **COD Collection Fee (`EGP`)**, and **Cash Settlement Cycle (`T+1` , `INSTANT` , `WEEKLY`)** across **ALL Partner Stores (`target_scope: "all"`)** or for a **Specific Partner Store (`target_scope: store_id`)**, plus bulk-settle all delivered sub-orders in one click.
- **FR-10 (Two-Way Live Customer Support Chat & 360° Visibility):** The Client Storefront must provide a **" 💬 Chat with Support"** drawer for shoppers to open support tickets (`SUP-xxxx`) and exchange live messages with the **Facilitator Admin Customer Support Desk**, while giving the Facilitator Admin full 360° visibility and override controls across all stores, inventory, deliveries, settlements, orders, and webhooks.

3.5 Client Order Monitoring, Historical Closed Archive & Tax Invoicing

- **FR-11 (Client Order Tracking & Milestone Pipeline):** Clients must be able to view their complete order history partitioned into **Active & In-Transit** (orders undergoing store packaging, courier pickup, or last-mile delivery) and **Completed & Closed** (delivered, cash-settled, or cancelled orders).
- **FR-12 (5-Stage Visual Stepper):** Each order card must display a real-time 5-stage progress stepper (`1. Order Placed` → `2. Payment Verified & Split` → `3. Store Packaged` → `4. Out for Delivery (Bosta 3PL)` → `5. Delivered & Settled`) with bilingual English/Arabic labels and completion timestamps.
- **FR-13 (Per-Store Manifest & Bosta AWB Transparency):** Orders fulfilled by multiple partner stores must display itemized sub-manifests, store origin governorates, packaging status, and direct clickable

Bosta waybills (`courier_tracking_id`).

- **FR-14 (Auditable & Printable Egyptian Commercial Tax Invoice):** Clients must be able to view and print official tax invoices (`INV-ORD-xxxx`) complying with **Egyptian E-Commerce Law No. 151/2020** and **VAT Law No. 67/2016**, detailing NileConnect Tax Reg No (`492-817-309`), Commercial Register (`184920`), itemized partner store breakdown, VAT amounts, and delivery charges.
- **FR-15 (Integrated Support Drawer & Cart Reorder):** Clients must be able to open a pre-filled support ticket referencing their specific order number with one click, or re-populate their cart with all order items to reorder seamlessly.

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4. Main Business Flowcharts

Flow 1: End-to-End Multi-Store Marketplace & Outsourced Delivery Lifecycle

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flowchart TD
    Client["Client Browses by Product Category on Web or iOS/Android App"] --> Cart["Adds Items from 1 or
    Cart --> SplitCalc["Engine Calculates Client Total, Facilitator Commission % & Net Store Payouts"]
    SplitCalc --> Reserve["Reserve Stock in Each Partner Store's Inventory & Split into Per-Store Sub-Orde

    Reserve --> PayChoice{"Client Selects Egyptian Payment Method"}
    PayChoice -->|"Paymob Split (Card/Meeza/Apple Pay/Wallet)"| Paymob["Paymob 3DS / Native SDK Approved (
    PayChoice -->|"valU BNPL (6/12/24m)"| ValU["valU OTP Approved"]
    PayChoice -->|"Fawry Kiosk Code (48h Hold)"| Fawry{"Paid at Kiosk Within 48h?"}
    PayChoice -->|"InstaPay QR / IPA"| Insta["Facilitator Admin Approves Receipt"]
    PayChoice -->|"Cash on Delivery (COD)"| WA["Client Confirms WhatsApp OTP (4455)"]

    Fawry -->|"Expired (48h TTL)"| Release["Cancel Order & Return Reserved Stock to Partner Store(s)"]
    Fawry -->|"Paid Cash at Kiosk"| Dispatch

    Paymob --> Dispatch["Auto-Generate Per-Store Bosta AWBs (Pickup: Store Branch -> Dropoff: Client)"]
    ValU --> Dispatch
    Insta --> Dispatch
    WA --> Dispatch

    Dispatch --> StorePack["Partner Store Packs Items in Store Portal -> Marks READY_FOR_COURIER_PICKUP"]
    StorePack --> CourierPickup["Outsourced Bosta Courier Picks Up Parcel from Partner Store Branch"]
    CourierPickup --> Deliver["Courier Delivers to Client & Collects COD (if applicable)"]
    Deliver --> Settle["Facilitator Keeps Commission (8-12%) & Settles Net Payout to Partner Store"]
    Settle --> Archive["Order Transitions to Closed Archive & Client Prints Tax Invoice"]
```

Flow 2: Financial & Commission Split Matrix

Order Component	Who Pays / Collects	Who Receives It	Example (10,000 EGP Item @ 10% Store Commission + 55 EGP Cairo Bosta)
Product Price Subtotal	Paid by **Client**	Split between **Partner Store** & **Facilitator**	`10,000 EGP`
Facilitator Commission (`8%–12%`)	Deducted from Subtotal	**Marketplace Facilitator (You)**	`1,000 EGP` *(Your Net Revenue)*
Net Partner Store Payout	Settled via Paymob Split / Bank	**Partner Store (Vendor)**	`9,000 EGP` *(Paid to Store)*
Outsourced Delivery Fee	Paid by **Client**	**3PL Courier (Bosta / Mylerz)**	`55 EGP` *(Covers Store-to-Client Pickup & Delivery)*

Flow 3: End-to-End Cash Settlement & Commission Realization (`STL-xxxxxx`)

1. **Eligibility Trigger (`ELIGIBLE_FOR_SETTLEMENT`):** A Partner Store sub-order becomes eligible for Cash Settlement only after the outsourced Bosta courier confirms `DELIVERED` to the Client.

2. **Prepaid Rail Settlement (`Paymob Split / Fawry / valU / InstaPay`):** Upon `POST /wp-json/wc/v3/deliveries/{id}/settle`, the escrow split releases `net_store_payout_egp` to the Partner Store's sub-merchant account and realizes `platform_commission_egp` into the Facilitator's revenue ledger (`STL-xxxxxx`).

3. **COD Rail Settlement (`Bosta COD Remittance -> InstaPay Corporate Payout`):** Bosta remits the collected `cod_to_collect_egp` cash; the Facilitator deducts its `platform_commission_egp` (`8%–12%`) and transfers `net_store_payout_egp` to the Partner Store via InstaPay Corporate / Bank Payout (`STL-xxxxxx`).

4. **Master Order Closure (`completed-settled`):** Once all Partner Store sub-deliveries for an order reach `SETTLED_TO_STORE`, the parent order automatically transitions to `order_status = 'completed-settled'` and `payment_status = 'settled'`.

5. **Historical Archive Partitioning:** Closed orders are preserved in the Client Order History archive with auditable financial summaries, delivery waybill history, and printable invoices.

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5. Non-Functional & Interactive UX Requirements

- **Eye-Friendly Adaptive Light/Dark Theme (`Plus Jakarta Sans` + `Cairo`):** Defaults to a high-contrast, low-glare **Light Theme** (`#f4f7fb` canvas with pure `#ffffff` cards, generous padding, and `14px` readable typography) with a 1-click 🌙 **Dark / 🌞 Light Mode Toggle** in the top header.
- **Focused Sub-Workspaces & Guided Step Pipelines (Zero Visual Clutter):**

- **Tab 1 (Client Storefront):** Fast category browsing, multi-store cart calculations, and 1-click shortcut to track active orders.
- **Tab 2 (My Orders & Tracking):** Dedicated shopper self-service hub with active vs. closed tabs, live search input, 5-stage visual stepper, and printable tax invoice modal.
- **Tab 3 (Mobile App Simulator):** iOS & Android mobile shopping experience with live Firebase push feed.
- **Tab 4 (Partner Store Portal):** Features 1-click **Visual Partner Store Cards** and prioritizes **Incoming Sub-Orders to Pack** at the top of the screen.
- **Tab 5 (Outsourced Delivery Hub):** Highlights the **Recommended Next Courier Step** (`Step 1: Store Packed` → `Step 2: Bosta Picked Up` → `Step 3: Out for Delivery` → `Step 4: Delivered` → `Step 5: Settle Cash`) as a single primary action button.
- **Tab 6 (Facilitator Control Tower):** Organizes the 360° Admin Console into **5 focused sub-tabs** (`1. Commercial Rules & Fees` , `2. Customer Support Chat Desk` , `3. Orders, Deliveries & Settlements` , `4. Inventory, Stores & Categories` , `5. Webhooks Inspector` , plus `Show All`) to eliminate scroll fatigue.
- **Event-Driven UI State Updates (No Destructive Polling):** All 6 workspaces update only upon user actions with instant visual Toast feedback (`showToast`), preserving open dropdowns and form inputs.
- **Dual-Stack (`IPv4 + IPv6`) Local & Vercel Cloud Parity:** Both the Core Engine (`:4000`) and Integrations Mock Server (`:4001`) run unchanged locally over dual-stack sockets AND deploy as a unified Vercel Python Serverless bundle (`vercel.json` + `api/index.py`) where `/` , `/wp-json/wc/v3/` , `/mock-dashboard` , `/mock/` , and `/study` preserve the exact same multi-tier REST + HMAC-SHA512 webhook architecture via an in-memory HTTP bridge (`invoke_handler_in_memory`) and ephemeral HPOS SQLite storage in `/tmp/ecommerce_hpos.db` .